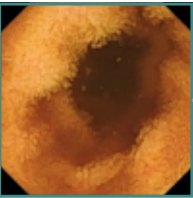


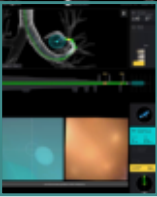
Lane A: passive-video forecasting
validated across three orifices



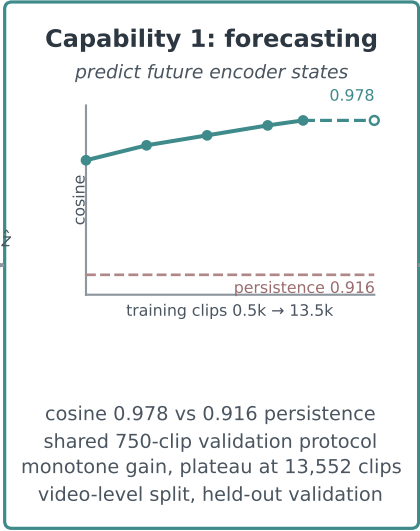
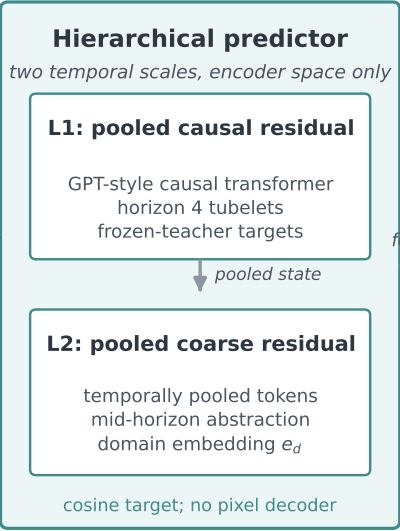
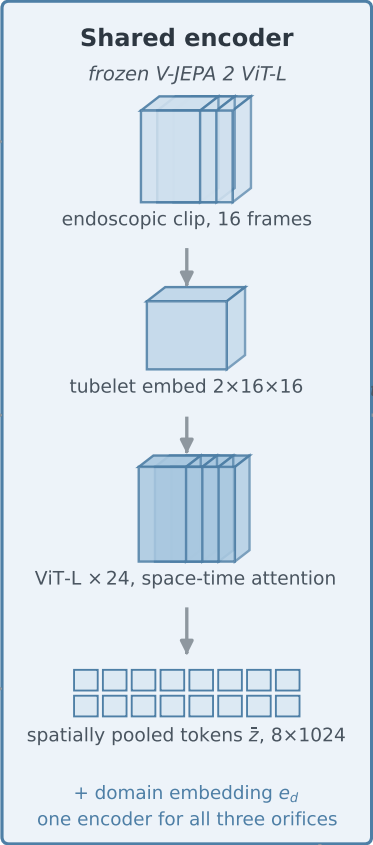
CholecT50 (laparoscopy)
last observed frame



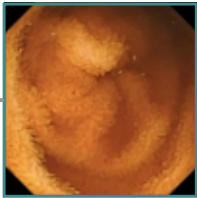
Kvasir-Capsule (GI)
last observed frame



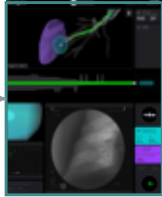
ION (bronchoscopy)
navigation-console capture



CholecT50 (laparoscopy)
predicted-latent retrieval

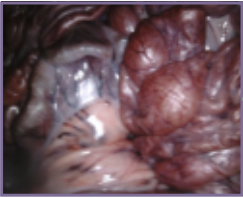


Kvasir-Capsule (GI)
predicted-latent retrieval

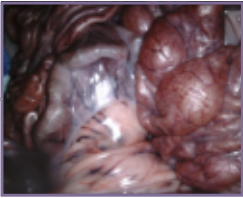
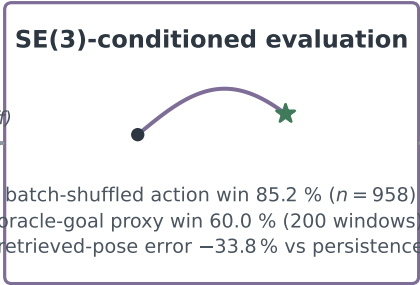
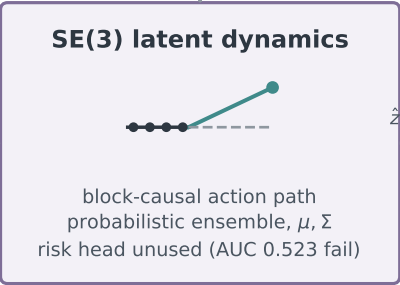
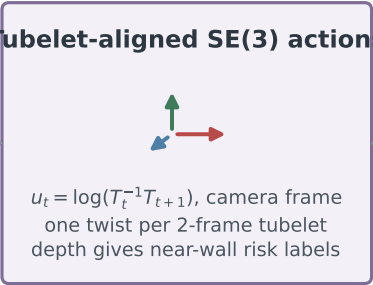


ION (bronchoscopy)
predicted-latent retrieval

Lane B: physical grounding
audited on pose-gated data only



SCARED (laparoscopy)
frame + calibrated pose



SCARED (laparoscopy)
SE(3)-conditioned retrieval

Correctness audits applied to both lanes

Input-sensitivity tests
history / action / domain

C3VD depth-warp diagnostic
convention not validated

Action negatives
batch shuffle / fixed bank

Grouped CV, logged test contacts
audit-selected result